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Key concept learning for medical vision language model with reasoning capabilities

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Training medical vision-language models (VLMs) typically demands millions of image-text pairs to achieve versatility and reasoning, posing significant challenges in data acquisition. We propose ConceptVLM, a novel data-efficient fine-tuning paradigm that transforms general-domain VLMs into specialized medical ones with minimal labeled data, integrating medical knowledge without disrupting the model's existing general capabilities. Central to our approach is a key concept-aware training strategy, building a structured medical concept dictionary and employing masked attention to guide the model's focus toward essential clinical concepts. This focused fine-tuning enhances domain-specific comprehension while preserving the model's reasoning abilities and response diversity. Experiments across multimodal medical benchmarks show ConceptVLM achieves state-of-the-art results using only 1% of the original training data, outperforming traditional methods reliant on large-scale QA datasets. These findings challenge the prevailing reliance on extensive annotated corpora, demonstrating key concept-guided tuning as a viable path to developing cognitively capable medical VLMs.

Vision-language models (VLMs), which integrate visual encoders with large language models (LLMs), hold transformative potential for medical image understanding and clinical decision support[1,2]. By interpreting medical images and responding to text-based queries, these models can function as interactive assistants for clinicians and patients, enhancing diagnostic accuracy and improving healthcare delivery[3–5].

Despite their promise, training multimodal medical VLMs presents significant challenges. Assembling a privacy-compliant, large-scale dataset with visual-language alignment is time-consuming and requires substantial clinical expertise, manual annotation effort, and institutional coordination. This greatly limits the feasibility of private customization, which is especially critical in medical scenarios due to stringent patient privacy requirements. Moreover, medical VLMs trained in this way underperform in multi-step reasoning, clinical judgment, and comprehension compared to general-purpose models in non-medical fields such as DeepSeek, GPT-4, and Qwen[6–8], largely because most medical QA data are templated and lack high-quality datasets capturing complex chains of thought. Currently, most of the medical VLMs[3,9] tend to provide only final answers or "quasi-explanations" derived from pre-training pattern matching without step-by-step reasoning. This limitation reduces model transparency and clinician trust, potentially leading to errors if the model's attention is not properly directed. Medical diagnosis often relies on subtle findings or specific regions in an image, for example, a tiny hemorrhage, drusen, or neovascularization may be decisive in diagnosing wet age-related macular degeneration in ophthalmology. Therefore, a key objective is to equip medical VLMs with the ability to focus on the most relevant visual evidence and articulate the reasoning behind their predictions, much like a human specialist would.

In this work, we propose ConceptVLM, a novel multi-modal medical question-answering framework that leverages key concept awareness to enhance its domain expertise (Fig. 1). Rather than following the conventional three-stage training, ConceptVLM directly fine-tunes the general-purpose VLM and incorporates domain knowledge through a high-quality QA corpus without compromising the generality and reasoning capabilities of the VLM. Note that domain-specific optimization can potentially degrade the general reasoning capacity and response diversity, and therefore we introduce a key concept-aware adaptation mechanism to overcome the catastrophic forgetting caused by limited and less diverse data fine-tuning.

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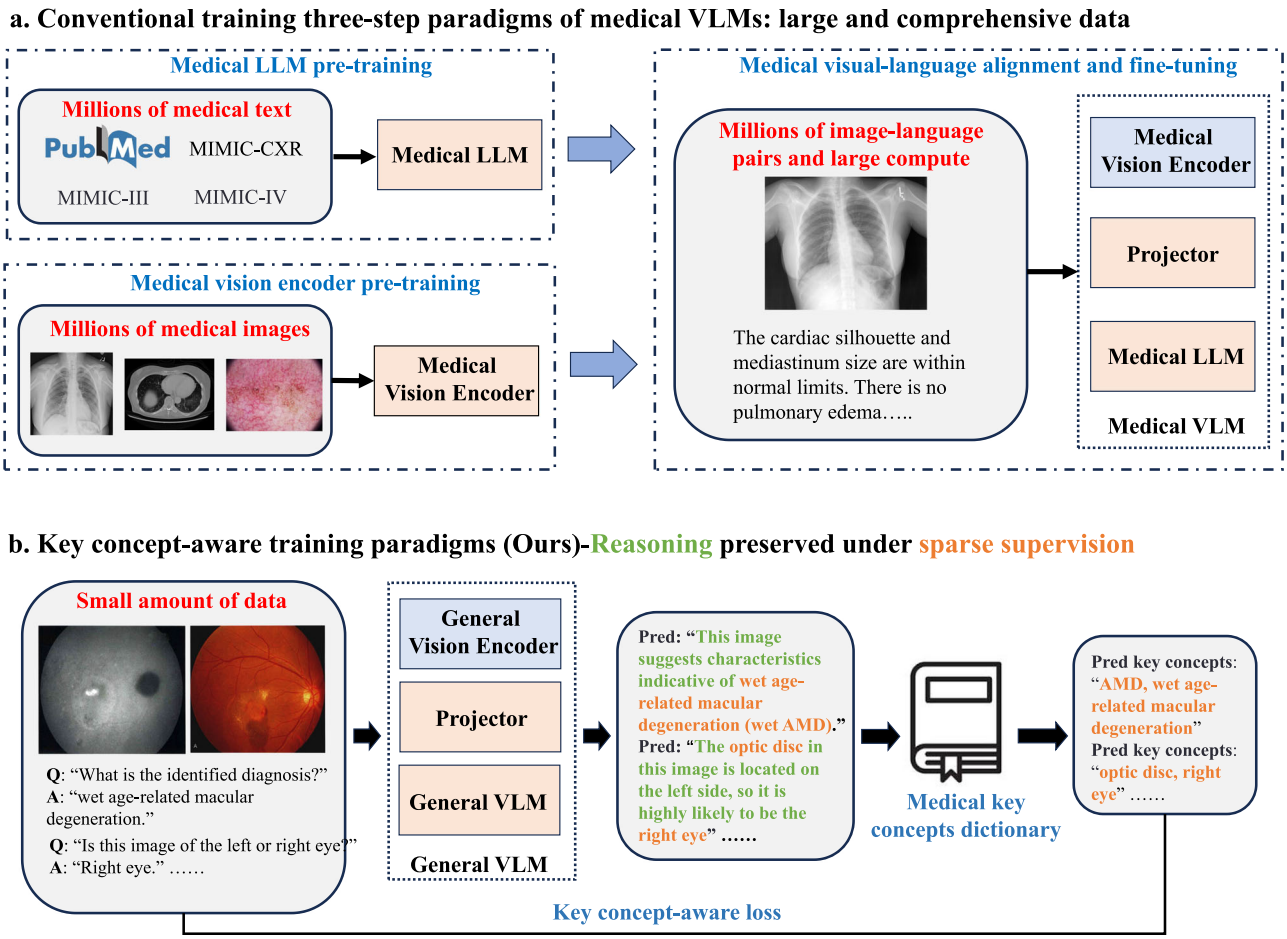


Fig. 1 | Motivation of ConceptVLM. a The conventional training paradigms of medical visual-language models consist of three parts: language model pre-training, vision encoder pre-training, and alignment and fine-tuning. It requires large amounts of data and large compute. **b** Our ConceptVLM employs a key concept-aware adaptation strategy, utilizing key concept matching and LoRA fine-tuning methods to fine-tune the general VLM. This approach aims to achieve strong performance and enhance reasoning ability with a small amount of data. LLM=large language model. VLM=vision-language model.

To establish such a pipeline for the medical question answering, first, we collect and construct an instruction-tuning dataset, including Slake[10], PathVQA[11], and PolyOphthalmology, a multimodal ophthalmic dataset with standardized captions and QA pairs. We then use this dataset to develop ConceptVLM and benchmark our proposed model on seven multimodal VQA datasets in ophthalmology, radiology, and pathology and four reasoning sets. Figure 2 shows the study design. Our results demonstrate that ConceptVLM achieves higher accuracy and better handles complex, multi-step questions compared to the baseline model, while utilizing merely 1% of the training data required by state-of-the-art counterparts. Notably, the proposed framework exhibits robust multitask proficiency in clinically relevant capabilities including radiology report generation, differential diagnosis rationalization, and interactive medical consultation without additional tuning. Overall, our approach demonstrates that it is feasible to construct a robust medical VLM with reasoning ability using low-cost, small-scale data, thereby providing possibilities for personalized clinical adaptation for a broad range of researchers and hospitals.

Results

Training dataset description

Unlike conventional training approaches for large foundation models that rely on extensive medical image-text instruction datasets, our method focuses on fine-tuning a general-purpose vision-language model (VLM) using a limited set of high-quality medical instruction data.

ConceptVLM employs three specialized medical vision-language datasets: Slake[10], PathVQA[11], and PolyOphthalmology. Slake is a public radiology VQA dataset comprising computed tomography (CT), magnetic resonance imaging (MRI), and radiography (X-ray) modalities, containing 9835 training question-answer (QA) pairs; PathVQA is a pathology image understanding dataset with 19,755 annotated QA pairs; PolyOphthalmology is a proprietary multimodal ophthalmology dataset curated by us, encompassing color fundus photography (CFP), optical coherence tomography (OCT), and fundus fluorescein angiography (FFA) imaging modalities paired with domain-specific clinical queries, totaling 13,536 training instances. The combined instruction-tuning corpus (43,126 QA pairs) represents a substantially more compact scale compared to conventional medical VLMs (Fig. 3c). Specifically, our training data volume aligns with LLaVA-med[3] (42k samples) while constituting merely 1% of RADFM[12] (around 3 M samples).

Architecture of ConceptVLM

Key concept-aware fine-tuning is the central innovation of this work. We start with a pre-trained vision-language model and adapt it to the medical QA domain by training on an instruction-tuning dataset. As shown in Fig. 2, the proposed framework comprises three key components. First, we leverage the pre-trained VLM[13], which has been trained on a broad spectrum of natural image-text pairs. Second, we introduce an algorithmic approach to construct a medical key concept dictionary, enabling the extraction of critical medical terms from Q&A or classification datasets. Third, we incorporate a key concept-aware loss function, ensuring precise responses without compromising the diversity of output formats.

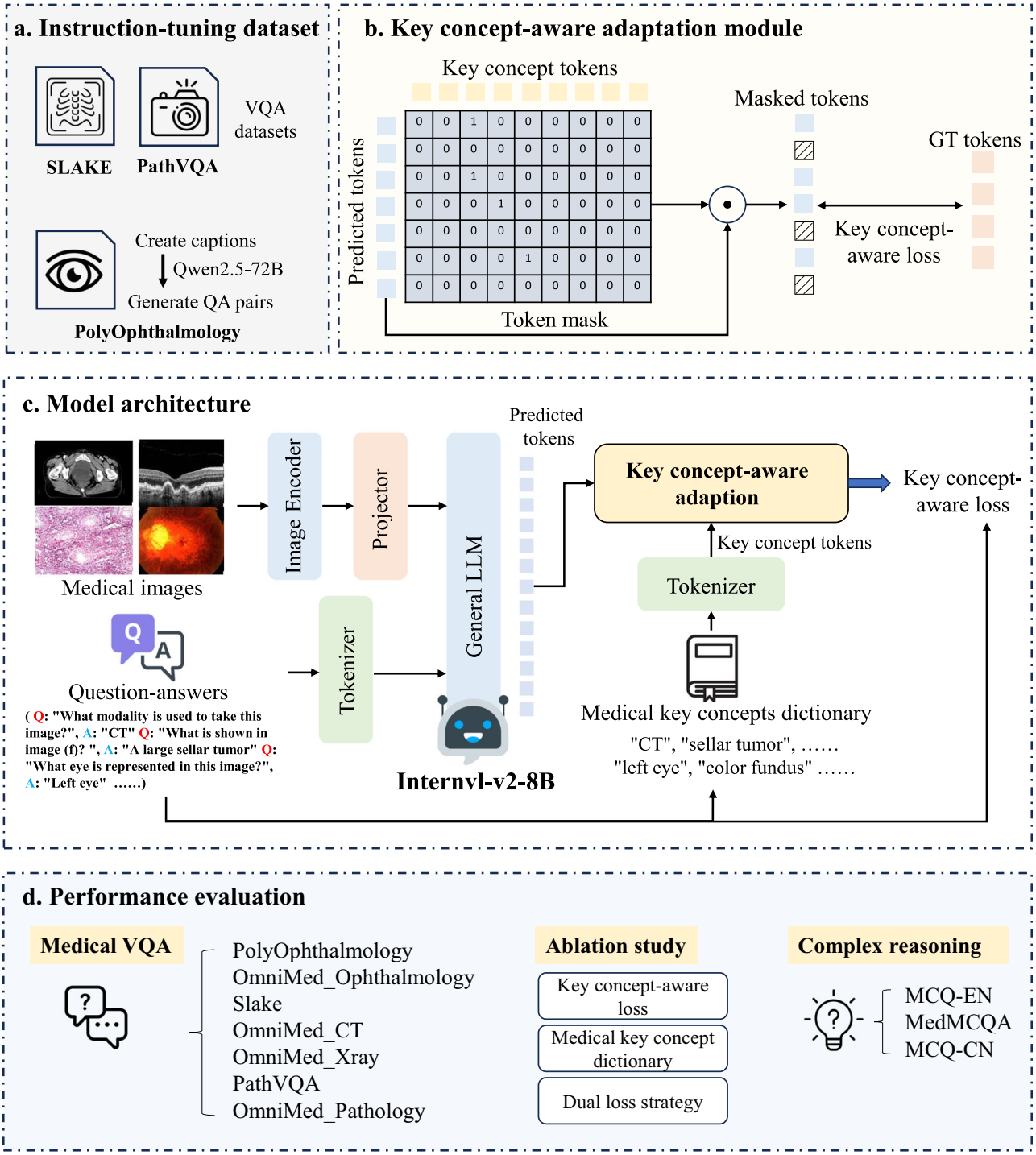


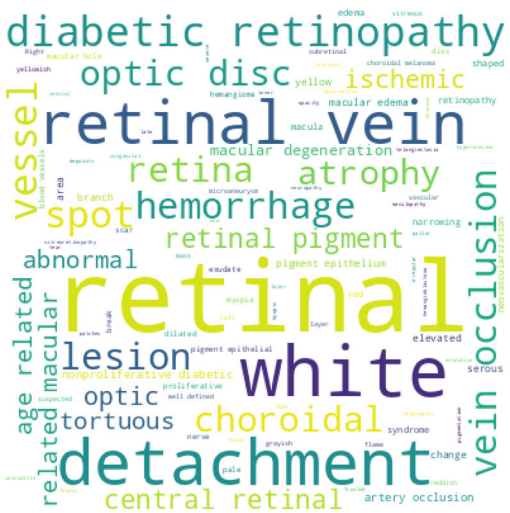
Fig. 2 | Study design of ConceptVLM. a Instruction-tuning dataset building. It contains three datasets: Slake, PathVQA, and PolyOphthalmology. **b** The key concept-aware adaptation module, which involves masking tokens and only computing the loss for tokens present in the medical key concept dictionary. **c** Overview of ConceptVLM architecture, where the overall framework includes an image encoder, a universal LLM, a tokenizer, and a medical key concept dictionary along with a key concept-aware loss. **d** Performance evaluation. ConceptVLM is evaluated on medical VQA tasks and complex reasoning tasks. An ablation study is used to isolate the effect of key concept-aware loss, medical key concept dictionary, and dual loss strategy. LLM = large language model. VLM = vision-language model. VQA = visual question-answers. QA = question-answers.

The VLM follows the "ViT-MLP-LLM" architecture, consisting of an image encoder, an MLP projector, and a large language model (LLM). Our approach diverges from traditional methods that compute cross-entropy loss directly between predicted and ground truth tokens. Instead, we construct a medical key concept dictionary and sample tokens from the LLM predictions that align with this dictionary before computing the loss. A token mask is created using the predicted tokens and the dictionary, setting the contribution of non-matching tokens to zero during loss computation. This ensures that only predictions matching the medical key concepts influence the loss, prioritizing medical relevance over format discrepancies and reducing overfitting.

a. Datasets used in our study

Dataset	Total	Modality
Training datasets		
Slake	9835 QA pairs	CT, MRI, X-ray
PathVQA	19755 QA pairs	Pathology
PolyOphthalmology	13536 QA pairs	CFP, OCT, FFA
Test datasets		
PolyOphthalmology	2096 QA pairs	CFP, OCT, FFA
OmniMed_Ophthalmology	6057 QA pairs	CFP, OCT
Slake	1062 QA pairs	CT, MRI, X-ray
OmniMed_CT	1980 QA pairs	CT
OmniMed_Xray	5974 QA pairs	X-ray
PathVQA	6719 QA pairs	Pathology
OmniMed_Pathology	2281 QA pairs	Pathology
MCQ-EN	139 QA pairs	Text
MCQ-CN	139 QA pairs	Text
MedMCQA_radiology	4395 QA pairs	Text
MedMCQA_pathology	14884 QA pairs	Text

b. Disease type / concept distribution



c. Instruction data usage and model performance

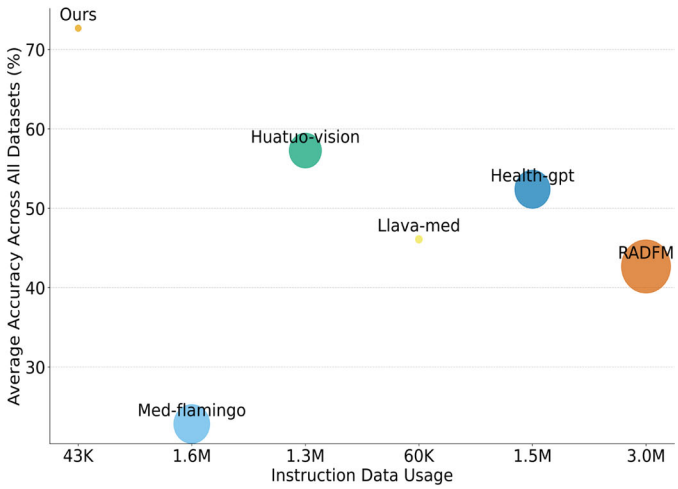


Fig. 3 | Statistics of the training and test datasets in our study. a Description of the data used in the training and testing sets. **b** A word cloud depicting medical concepts extracted from the training data, with word size proportional to its frequency of occurrence. **c** Comparative performance analysis contrasting the performance of our ConceptVLM and state-of-the-art methods on various benchmarks.

Additionally, we differentiate between closed-ended and open-ended question-and-answer (Q&A) data in our training process. For closed-ended Q&A data, characterized by short, structured answers, we apply the key concept-aware loss across all samples. For open-ended Q&A data, which exhibit diverse formats and cover multiple key concept categories, constructing a unified dictionary is challenging. To address this, we introduce a length threshold criterion: if the ground truth answer is shorter than the threshold and contains at least one concept from the medical key concept dictionary, we apply the key concept-aware loss; otherwise, we revert to conventional cross-entropy loss. This dual-strategy approach enables the model to effectively handle both structured simplicity and unstructured complexity.

Multi-modal visual question answering

We evaluated the visual question answering performance of ConceptVLM on multiple benchmarks across various modalities. These benchmarks included not only the in-domain test sets from radiology dataset Slake[10], pathology dataset PathVQA[11], and our private dataset PolyOphthalmology,

but also out-of-domain (OOD) datasets that lie outside the training data domains. The OOD datasets were primarily constructed by selecting subsets from the large-scale OmniMedVQA benchmark[9] according to modalities, comprising the Omnimed_Pathology dataset, Omnimed_CT dataset, Omnimed_Ophthalmology dataset, and Omnimed_Xray dataset. Figure 4a, b summarize the results, comparing our model to multiple state-of-the-art methods: (1) Internvl-v2-8B[13], the baseline general-purpose VLM model without concept fine-tuning, and (2) existing specialized medical VLMs: Huatuo-vision[14], Llava-med[3], RadFM[12], Med-flamingo[15], and Health-GPT[16]. For Huatuo-vision, we utilized the 7B version, and for Health-GPT, we employed the Health-GPT-L14 model. We also demonstrate the rich medical question-answering capabilities of our model, providing visual examples in Fig. 5a and Supplementary Fig. 1.

Overall, ConceptVLM achieved a higher overall accuracy than the baseline Internvl-v2 in all the datasets. Specifically, ConceptVLM improved Internvl-v2 by 17.9% on Omnimed_Ophthalmology, 2.5% on Omnimed_CT, 2.7% on Omnimed_Pathology, and 5.9% on Omnimed_Xray dataset. Notably, the advantage of concept-aware fine-tuning was most pronounced on the open-ended questions. In some cases, questions requiring the identification of a specific abnormality or the description of a finding were answered correctly by ConceptVLM, whereas the baseline often produced incomplete or incorrect answers (Fig. 5b). While closed-ended questions were already reasonably well addressed by the baseline model, ConceptVLM further reduced the error rate, indicating that concept guidance helps the model focus on the relevant visual features when making decisions.

Ablation results further demonstrate that each design (key concept-aware loss, medical key concept dictionary and dual loss strategy) effectively improves the accuracy of medical question answering (Fig. 4c). The 'Baseline' in Fig. 4c refers to Internvl-v2-8B fine-tuned using standard LoRA with full answer supervision. The medical key concept dictionary plays a foundational role by distilling clinically salient terms from training data, allowing the model to prioritize diagnostically relevant information rather than learning concepts implicitly from sparse QA pairs. The key concept-aware loss builds on this by selectively supervising only those tokens associated with entries in the dictionary, which sharpens attention to critical visual evidence while mitigating overfitting and preserving general reasoning capabilities during low-data adaptation. Meanwhile, the dual loss strategy handles the diversity of answer formats: it applies concept-guided supervision to short, structured responses where medical precision is essential, and reverts to standard cross-entropy loss for longer, open-ended answers to maintain linguistic fluency. The necessity of this dual loss strategy is confirmed in a ablation (Supplementary Table 12), which shows that relying solely on the key concept loss for all answer types degrades performance on open-ended questions. Notably, the construction of the medical key concept dictionary significantly enhances the model's focus on medical knowledge, leading to accuracy improvements of 6.1%, 4.6%, and 4.5% on Slake, PathVQA, and PolyOphthalmology, respectively. Importantly, qualitative analysis confirms that while the dictionary guides attention, it does not restrict the model's vocabulary. As illustrated in Supplementary Fig. 6, ConceptVLM maintains high semantic coherence and accurately generates complex clinical descriptions even when specific key medical concepts are absent from the concept dictionary (highlighted in yellow). For instance, the model dynamically formulates precise pathological signs such as "neuroretinal rim loss" in a glaucoma case (Supplementary Fig. 6b) and correlates "punched-out areas" with "multiple myeloma" in gross pathology (Supplementary Fig. 6e). Moreover, to verify the generality of our method, we have added comparative experiments with two different backbones to evaluate the compatibility of our framework with stronger backbones. Specifically, as shown in Supplementary Table 7, we adopted the Qwen2.5-vl-7B[17] model, which outperforms Internvl-v2-8B on two out-of-domain datasets. The experimental results demonstrate that our approach achieves a 5.8% improvement on Omnimed_Ophthalmology and a 1.6% improvement on Omnimed_Xray. Therefore, our training framework effectively enhances the model's question-answering accuracy when combined with two different backbone architectures. To assess the effectiveness of concept-guided supervision, we also compared ConceptVLM with a Direct Preference Optimization (DPO) reinforcement learning baseline on PolyOphthalmology test set (Supplementary Table 11). Our method outperforms DPO across all metrics—by +4.4% in recall, +9.6% in accuracy, and +3.6% in ROUGE-L.

For the in-domain dataset, ConceptVLM achieved a mean accuracy of 76.7%, significantly outperforming the previous state-of-the-art method, Huatuo-vision's 32.4% on the PolyOphthalmology dataset ($P < 0.001$). On Slake and PathVQA datasets, our ConceptVLM achieved a mean accuracy of 90.4% and 65.6%, respectively, surpassing the second-best methods Llava-med by 3.5% ($P < 0.01$) and 0.3% ($P > 0.05$) (Fig. 4a, b, Supplementary Table 1). For the OOD datasets, ConceptVLM outperforms the second-best models, Huatuo-vision by 1.2% ($P < 0.05$) on the Omnimed_Pathology dataset. On the Omnimed_Ophthalmology datasets, ConceptVLM achieved 70.5% accuracy, which surpasses Huatuo-vision's 67.2% by 3.3% ($P < 0.01$). In addition, ConceptVLM shows similar performance to the current leading method, Huatuo-vision, with a difference of less than 0.5% ($P > 0.05$) on the Omnimed_CT and Omnimed_Xray datasets (Fig. 4d). Given the smaller amount of training data used, we believe this minor gap is acceptable. To further validate the necessity of the concept-aware strategy for cross-domain generalization, we compared ConceptVLM against a standard LoRA-fine-tuned Internvl-v2 baseline and a standard LoRA-fine-tuned Huatuo-vision model (Supplementary Table 10). The results reveal that standard fine-tuning on a specialized dataset without concept guidance leads to "catastrophic forgetting," where the model overfits to the training domain and suffers performance degradation on OOD tasks such as radiology and pathology. In contrast, our concept-aware strategy not only prevents this decline but actively facilitates positive transfer, resulting in significant improvements over the baseline model (e.g., +4.2% on Omnimed_CT and +3.7% on Omnimed_Pathology).

We also evaluated the accuracy of various methods on 'open-ended' and 'closed-ended' question-answer pairs accordingly. ConceptVLM demonstrates robust performance in both open-ended and closed-ended question answering tasks on the Slake dataset, achieving improvements of 3.3% and 3.6%, respectively, over the second-best method, LLaVA-med. On the PathVQA dataset, ConceptVLM shows a 1.5% improvement in closed-ended question answering accuracy compared to the second-best method. Furthermore, on the PolyOphthalmology dataset, ConceptVLM significantly outperforms previous state-of-the-art methods by over 20% in both open-ended and closed-ended question-answering tasks ($P < 0.001$). These quantitative results validate our approach: explicitly teaching the model about key concepts yields tangible gains in multi-modal VQA ability.

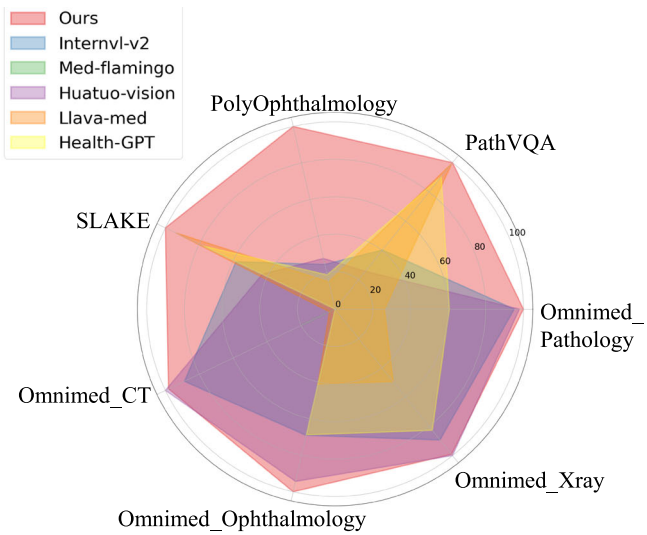
Complex reasoning ability of ConceptVLM

We evaluated the complex reasoning capabilities of ConceptVLM on five complex reasoning and clinical generation datasets: MCQs_CN[18], MCQs_EN[18], MedMCQA_radiology[19], MedMCQA_pathology[19] and MIMIC-CXR-RRG[20]. Among these, MCQs_CN consists of ophthalmology questions in Chinese, while MCQs_EN contains ophthalmology questions in English. As shown in Fig. 6a and Supplementary Fig. 2 and Supplementary Fig. 3, ConceptVLM can provide detailed analyses of the questions and options while giving answers, demonstrating its ability to perform logical reasoning. The numerical results in Fig. 6b and Supplementary Table 3 also clearly indicate that our training approach effectively enhances the reasoning capability of the baseline model, Internvl-v2 (all $P < 0.05$). Additionally, our model achieves an accuracy of 0.692 on the Chinese reasoning dataset MCQs_CN, reaching a level comparable to powerful general reasoning models such as OpenAI o1 and OpenAI o3-mini ($P > 0.05$). However, regarding English reasoning capabilities, our ConceptVLM still lags behind general reasoning models, limited by the capacity of the baseline model. To deeply understand these cross-lingual performance variations, we conducted a granular qualitative error analysis (Supplementary Figs. 7, 8). We observed that the

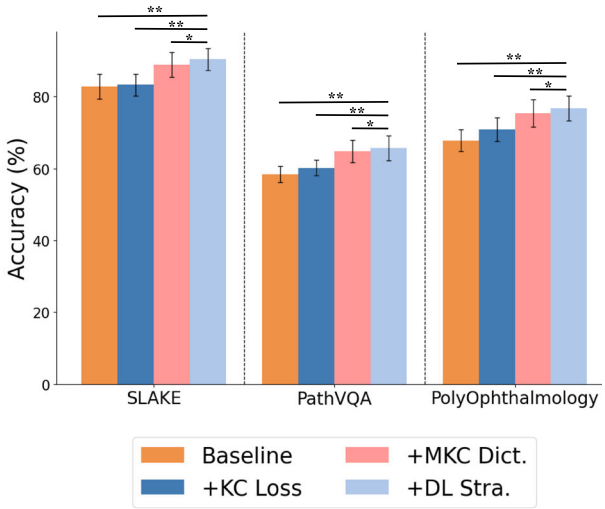
a. VQA performance on Open/Close-ended questions

Model	SLAKE			PathVQA			PolyOphthalmology		
	Open	Close	Mean	Open	Close	Mean	Open	Close	Mean
Internvl-v2	62.5	71.8	67.2	20.5	65.1	42.8	11.2	38.8	25.0
Llava-med	87.1	86.7	86.9	**39.6**	91.1	65.3	35.7	13.4	24.6
Med-flamingo	23.5	45.7	34.6	2.1	52.7	27.4	9.0	4.1	6.6
Huatuo-vision	41.0	76.0	58.5	8.4	66.1	37.3	51.1	13.7	32.4
Health-GPT	78.4	77.2	77.8	32.4	91.3	61.9	50.3	3.1	26.9
Ours	**90.4**	**90.3**	**90.4**	38.5	**92.6**	**65.6**	**72.7**	**80.7**	**76.7**

b. Medical VQA performance



c. Ablation study of VQA performance



d. VQA performance on OOD datasets

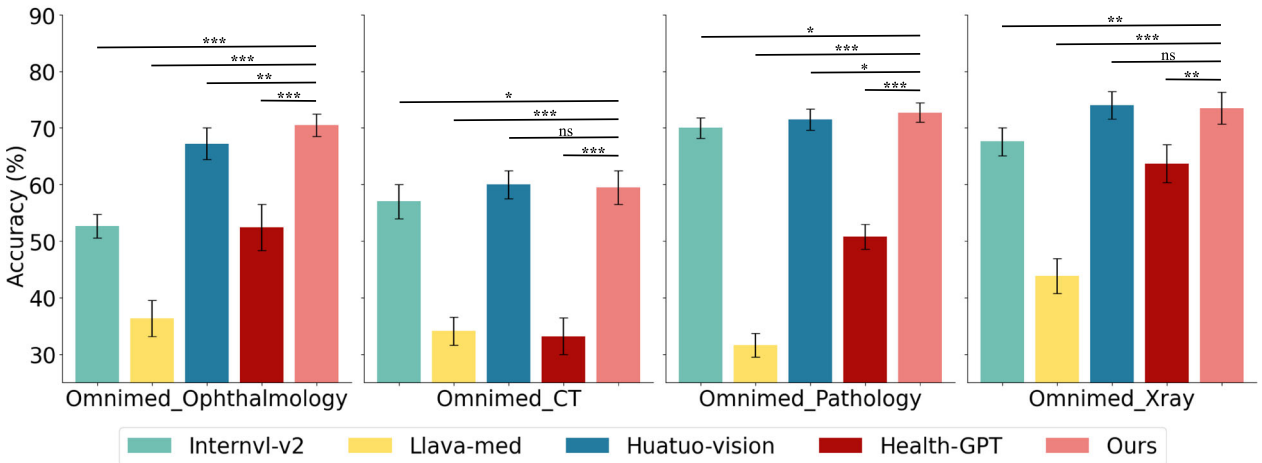


Fig. 4 | Medical VQA performance. a Medical VQA performance of ConceptVLM and the leading models on three medical datasets, in terms of closed-ended accuracy and open-ended recall. **b** Comparative performance analysis contrasting the performance of our ConceptVLM and state-of-the-art methods on various benchmarks. **c** Ablation study on three datasets. The "Baseline" represents the direct LoRA fine-tuning of the baseline model (Internvl-v2). "+ KC Loss" indicates that the baseline model is trained using the key concept-aware loss. "+ MKC Dict." means that keywords are first matched using a medical key concept dictionary, followed by training with key concept-aware loss. "+ DL Stra." refers to the use of a dual loss strategy on top of the "+ MKC Dict." model to differentiate losses for open-ended question answering. **d** Medical VQA performance on out-of-domain (OOD) datasets. The results show that our method can achieve on-par performance with the existing state-of-the-art (SOTA) methods on CT and X-ray data modalities and surpass the existing SOTA methods on pathology and ophthalmology modalities.

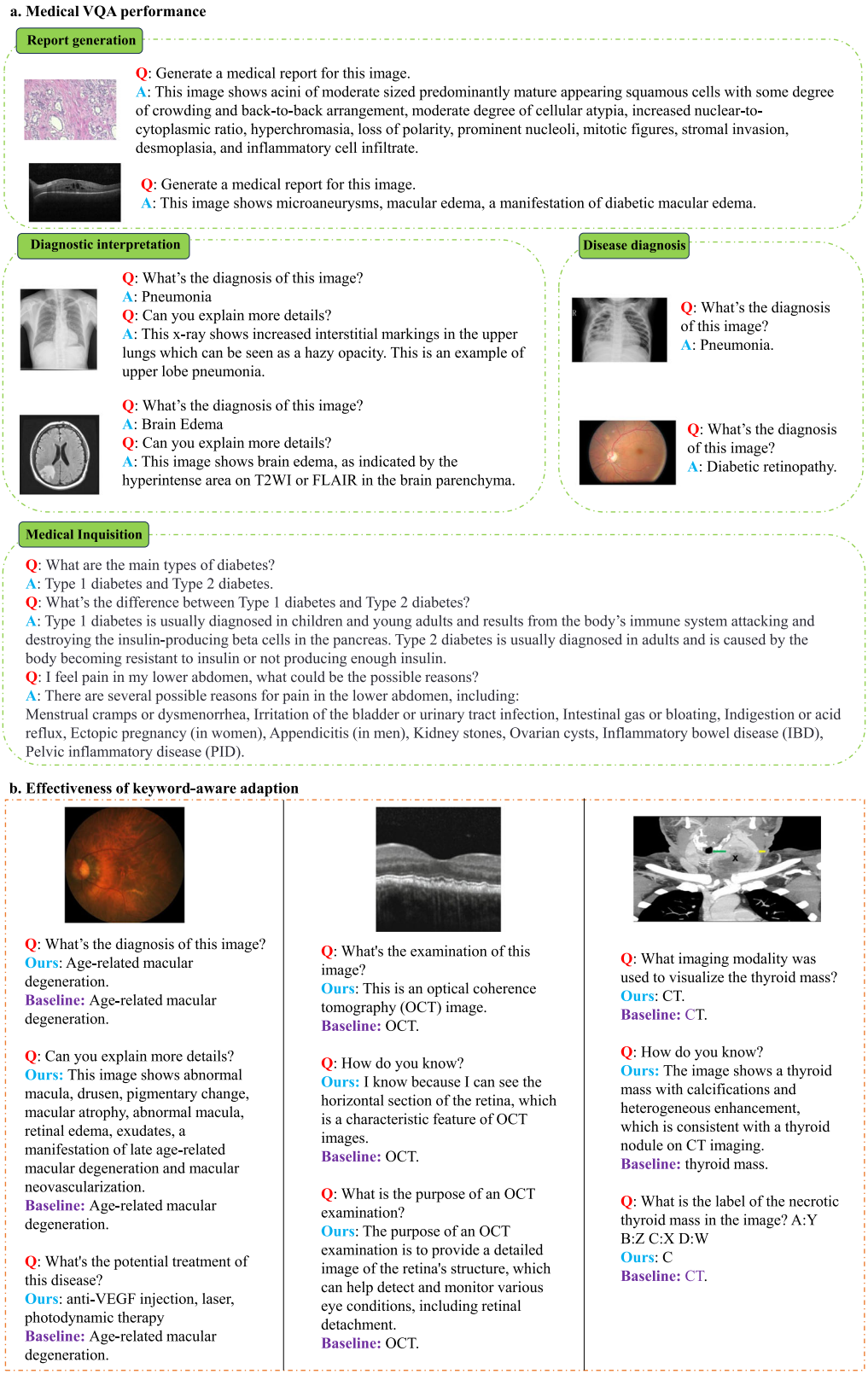


Fig. 5 | ConceptVLM can perform well on multiple modalities and tasks. **a** Examples of use cases like report generation, disease diagnosis, diagnosis interpretation, and image-free medical inquiry. "Q" denotes a question of the users, and "A" is the answer of ConceptVLM. With the help of the key concept-aware adaptation mechanism, ConceptVLM is capable of comprehensively and accurately handling different types of question-answer tasks. **b** Comparison between a baseline model ("Baseline"), which directly fine-tunes on the same universal foundation model, and our proposed ConceptVLM ("Ours"). The results show that using a key concept-aware adaptation mechanism can effectively retain the analytical capabilities of the universal large model, whereas direct fine-tuning leads to degradation of general capabilities, causing the model to only answer questions related to our training data and losing its generalizability.

a. Complex reasoning ability

Question: Female, presents with left eye redness, swelling, and increased discharge for 1 day. Examination reveals left eyelid erythema and edema, copious yellow-white discharge in the conjunctival sac (recurring upon wiping), conjunctival congestion and chemosis, tender preauricular lymphadenopathy, and clear cornea. Accompanied by symptoms of urinary frequency, urgency, and dysuria.
[Single-choice question]
What is the most likely diagnosis?
A. Vernal keratoconjunctivitis
B. Epidemic keratoconjunctivitis
C. Gonococcal conjunctivitis
D. Trachoma
E. Acute conjunctivitis
Reference answer: C
Our answer: C.
Vernal keratoconjunctivitis is characterized by itchy eyes, clear mucoid discharge, and papillary conjunctivitis. Epidemic keratoconjunctivitis is caused by adenovirus and presents with acute onset of conjunctival injection, chemosis, and watery discharge. Trachoma is caused by Chlamydia trachomatis and presents with conjunctival injection, follicular keratitis, and lymphadenopathy. Gonococcal conjunctivitis is caused by Neisseria gonorrhoeae and presents with purulent discharge, chemosis, and tender preauricular lymphadenopathy. Acute conjunctivitis is caused by adenovirus and presents with acute onset of conjunctival injection, chemosis, and watery discharge.

b. Complex reasoning accuracy comparison

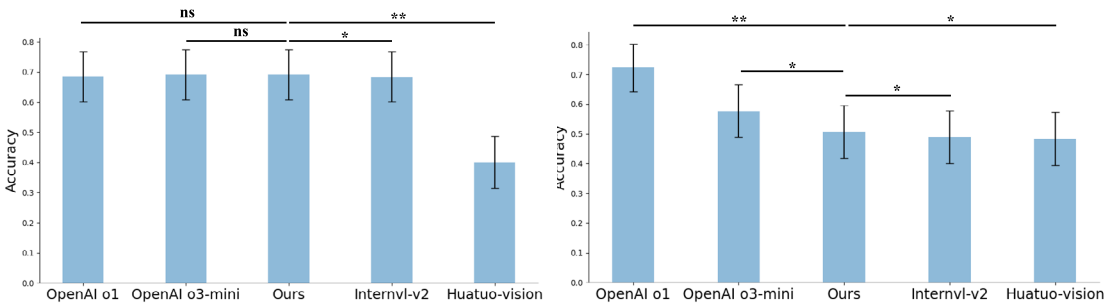


Fig. 6 | Complex reasoning performance of ConceptVLM. a Given complex reasoning problems, ConceptVLM can provide a detailed analysis and reasoning process, along with a final decision. **b** We compared the reasoning capabilities of ConceptVLM with current mainstream multimodal reasoning models and medical visual-language models on the MCQs_CN and MCQs_EN datasets. The results show that our key concept-aware adaptation paradigm improves the complex reasoning ability of the Internvl-v2 model and achieves performance comparable to the current mainstream multimodal reasoning models in complex reasoning tasks.

model's failure modes are highly language-dependent: the English model often suffers from "single-choice collapse" (ignoring multi-select instructions) and flawed exclusion logic, whereas the Chinese model tends towards "defensive over-selection" (sacrificing precision for recall) and false causal anchoring on historical keywords. These distinct heuristic biases highlight the inherent complexities of multilingual medical alignment beyond mere domain knowledge. For MedMCQA_radiology and MedMCQA_pathology datasets, we further provide experiments in Supplementary Table 8 to validate the reasoning capabilities of ConceptVLM on radiology and pathology questions. Experimental results show that our method enhances Internvl-v2's reasoning capabilities in radiology and pathology, while also surpassing a state-of-the-art medical vision-language model. For MIMIC-CXR-RRG dataset, we evaluated the zero-shot report generation performance in Supplementary Table 9. Since ConceptVLM was not explicitly fine-tuned on large-scale report templates, it maintained competitive, albeit statistically comparable, performance across exact-match metrics (BLEU, METEOR, $p > 0.05$ compared to baselines). Importantly, however, it achieved a statistically significant improvement in the ROUGE-L metric ($p < 0.05$). Because ROUGE-L evaluates the longest common subsequence, this suggests that our concept-guided tuning helps preserve and slightly enhance the structural coherence and logical sequencing of clinical narratives, demonstrating the method's potential to generalize beyond short-form VQA.

Data efficiency of ConceptVLM

To evaluate the data efficiency of our ConceptVLM framework, we conducted controlled experiments using progressively increasing proportions (10%-40%) of training data on the SLAKE and PathVQA benchmarks. As shown in Fig. 7 and Supplementary Table 2, our model demonstrates remarkable sample efficiency compared to baseline models: with only 10% training data, ConceptVLM outperforms the general-domain Internvl-v2-8B on open-ended questions across both datasets. At 40% data utilization, ConceptVLM surpasses the current state-of-the-art medical QA model LLaVA-med, achieving superior performance on both open-ended and closed-ended question types in the SLAKE and PathVQA test sets. This comparative analysis substantiates the exceptional capability of ConceptVLM in extracting maximum information value from limited training samples.

Discussion

In this study, we propose a novel paradigm for constructing large medical question-answering models: efficiently fine-tuning a general-purpose foundational VLM to a specialized one by constraining the optimization objective through a key concept-aware adaptation mechanism. This approach enables the injection of critical medical information while preserving the generality and reasoning capabilities of the VLM. Our training strategy is specifically designed to address the data scarcity and limited task diversity commonly found in medical settings, achieving strong

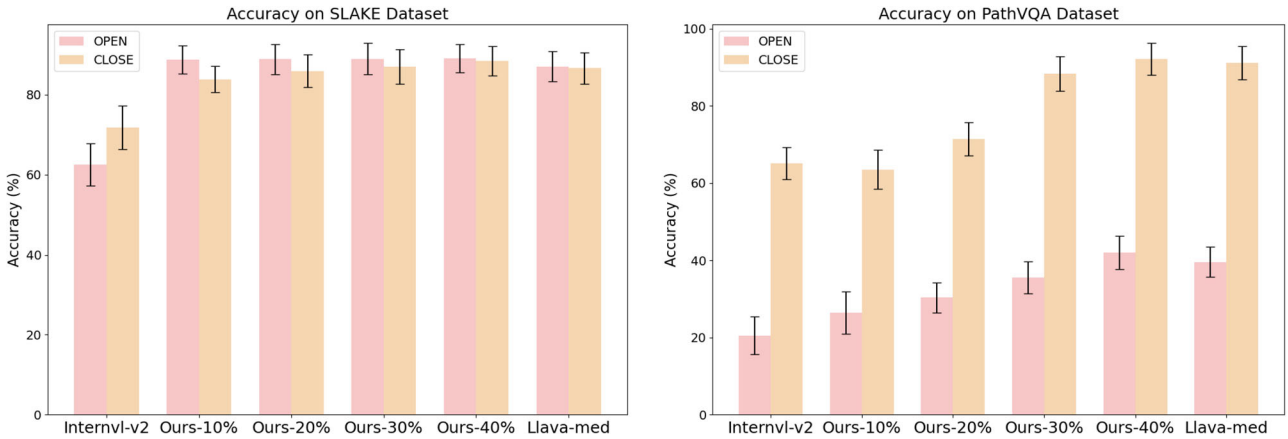


Fig. 7 | Performance comparison of ConceptVLM model using different proportions of instruction data. The results show that ConceptVLM can achieve good performance with a relatively small proportion of data.

generalization and reasoning capabilities using only simple classification datasets or monotonous question-answering datasets.

A major advantage of ConceptVLM lies in its ability to efficiently utilize small amounts of data, as demonstrated in our multimodal VQA tasks and comparisons across different ratios of training samples. In contrast, traditional medical VLMs typically require a large volume of instruction data and aim to diversify the forms of question-answering to ensure the model exhibits diversity and good generalization[21]. The difficulties in dataset construction significantly limit the feasibility of privatizing medical VLMs. Given the sensitivity of medical data, models that can be privately deployed at a low cost are essential. Therefore, our proposed training strategy provides a key medical concept-based training mode, which leverages the inherent knowledge of general-purpose VLMs and adapts them to private data, greatly reducing the requirements for training data.

Another key finding of ConceptVLM is that injecting domain-specific knowledge in the form of key concepts can substantially improve a general-purpose VLM's performance and reasoning capabilities within the medical domain. Importantly, our framework differs fundamentally from both conventional masking strategies and selective supervision strategies. The conventional masking models mask input tokens or patches and trains the model to reconstruct them via self-supervision, our method operates during supervised fine-tuning and involves no input masking or content reconstruction. Instead, it selectively computes the training loss only over output tokens that match clinically salient concepts from a curated dictionary. Moreover, while typical selective supervision relies on heuristic criteria (e.g., token position, answer length, or model confidence) to downweight predictions, our selection is grounded in domain semantics, making it a knowledge-guided mechanism rather than a generic efficiency trick. This approach differs from most existing adaptation techniques for large multimodal models[22]. Previous works on medical VLMs often focused on either pre-training on medical image-text data or fine-tuning with supervised QA pairs alone. While those methods capture correlations between images and answers, they do not explicitly teach the model how to reason about the question. In ConceptVLM, by contrast, we guide the model to consider "What is the question really asking?" through the concept prediction task. We validated the improved reasoning capability of ConceptVLM compared to the baseline model in medical complex reasoning tasks. This small addition has an outsized effect: it acts as an implicit reasoning step inserted into the training process. The model learns to break down the question into a core concept, mirroring how a clinician would approach a problem by identifying the relevant finding or condition before formulating an answer. Essentially, we align the model's latent reasoning by providing domain-specific simple question-answering data, thus avoiding the need for expensive manual annotations of full reasoning chains. In contrast to conventional supervised fine-tuning that applies uniform loss over all answer tokens, our approach focuses learning exclusively on high-value clinical concepts. By doing so, it not only avoids overfitting to superficial answer patterns but also implicitly preserves general-domain knowledge, which manifests as retained response diversity even under extreme data scarcity. This makes the construction of multi-task VLMs much simpler.

Additionally, ConceptVLM demonstrates good domain generalization, achieving performance improvements on test sets outside the training data. The concept-aware approach essentially injects domain knowledge into the model's training objective. We are leveraging the fact that the model does not need to discover from scratch which visual features matter, and it is pointed to the likely relevant feature. This guided learning is especially useful in medicine, where some findings can be subtle or easily overlooked by a general model. Our approach shows that the key concept-aware adaptation could not only be effective for question-answering tasks included in the training data but also indirectly enhances the model's sensitivity to medical keywords. Even medical keywords not present in the training data exhibit a degree of probabilistic enhancement, leading to improved accuracy in answering out-of-domain questions. This indicates that the concept-aware approach serves as an effective means of incorporating expert knowledge, particularly in scenarios with limited data availability. The mechanism is analogous to instructing the model during training to "focus on specific concepts in the image," which proves especially valuable for addressing rare or complex clinical cases.

Moreover, ConceptVLM is more efficient and cost-effective. Many existing approaches to enhancing reasoning capabilities involve heavy resource consumption and depend on refined feedback data obtained through reinforcement or active learning processes, while others rely on extensive chain-of-thought datasets for supervised training—both of which result in significantly higher costs. By comparison, our concept-aware adaptation training employs essential medical concepts to steer general-purpose vision-language models toward more accurate reasoning and responses. This strategy minimizes the reliance on intensive data collection and large-scale supervision, providing a streamlined and affordable solution. Furthermore, preliminary results on radiology report generation (Supplementary Table 9) suggest our framework offers a promising direction for mitigating structural data limitations in broader clinical tasks. The statistically significant improvement in ROUGE-L points to enhanced logical flow and structural coherence. While VQA remains our primary focus, this initial evidence of transferability suggests that grounding models in key concepts via small-scale data could be a helpful approach for extending medical VLMs to longer descriptive tasks.

Our study has several limitations. First, despite the improved response flexibility, some failure cases still exhibit misdiagnosis or overly simplistic answers, primarily due to limited data quality and pathological coverage. Second, our training paradigm relies on the capabilities of existing universal VLMs. For example, in English medical complex reasoning question-answering experiments, although our method effectively improves the

baseline model's performance, it still falls short of the best reasoning models. More importantly, our method inherits any weaknesses of the base model, such as poor image-text alignment or limited clinical knowledge, which underscores the need for foundation models better suited to healthcare. Third, due to the lack of multimodal medical knowledge, our current method for constructing the key concept dictionary is relatively rudimentary, relying on deduplication of answers and length-based filtering, which may introduce noise. Finally, all evaluations are conducted on retrospective benchmark datasets under controlled settings. We have not yet tested the system in real clinical workflows, where factors like time constraints, user interaction patterns, and integration with electronic health records would critically influence its utility and safety. In future work, we will evaluate the training paradigm on more capable foundation models, test the dictionary derived from structured medical ontologies, explore retrieval-augmented reasoning to dynamically access external knowledge, and collaborate with clinicians to conduct human–AI co-evaluation studies in realistic care environments.

Methods

Data curation and preprocessing for training

To construct the PolyOphthalmology dataset, we first collected 22 publicly available ophthalmology datasets, comprising a total of 73k images. To construct image captions, we standardized all captions using our self-developed condition standardized dictionary, creating captions that include information on laterality, modality, diagnosis, and lesion details, resulting in a total of 4,249 caption pairs. Subsequently, we used the Qwen2.5-72B model[17] to generate question-answer pairs from these captions, using the instruction prompts specified in Supplementary Fig. 5. These question-answer pairs were then filtered through human validation: two ophthalmology-trained annotators independently reviewed each pair for clinical relevance, factual accuracy, and alignment with the source caption. Pairs with ambiguous, incorrect, or hallucinated content were discarded. Ultimately, we sampled 13,536 QA pairs for the training set, including 1853 on modality, 2000 on laterality, 7023 on diagnosis, and 2660 open-ended image description pairs.

Data curation for downstream validation

To evaluate the performance of ConceptVLM in multi-modal visual question answering, we assessed it across seven datasets covering ophthalmology, radiology, and pathology modalities.

Ophthalmology validation datasets. We constructed two ophthalmology validation datasets, one is the PolyOphthalmology dataset[23] and the other is Omnimed_Ophthalmology dataset. The PolyOphthalmology test set consists of 2096 QA pairs in total, including 397 laterality QA pairs, 415 modality QA pairs, 490 diagnosis QA pairs, and 794 open-ended image description QA pairs. The Omnimed_Ophthalmology dataset was constructed based on eight publicly available non-VQA datasets (ACRIMA[24], Adam Challenge[25], Diabetic Retinopathy[26], DRIMDB[27], JSIEC[28], OCT & X-Ray 2017[29], OLIVES[30], Eye-Laterality[31]), totaling 6057 QA pairs.

Radiology validation datasets. For radiology, three evaluation datasets were collected: the test set of Slake[10] dataset, Omnimed_CT dataset and Omnimed_Xray dataset. The test set of Slake dataset includes 1062 QA pairs of CT, MRI and X-ray images. The Omnimed_CT dataset combines three public datasets (Chest CT Scan[32], Covid CT[33], SARS-CoV-2 CT-scan[34]) with a total of 1,980 QA pairs focusing on CT images. The Omnimed_Xray dataset encompasses eight datasets (Chest X-Ray PA[35], CoronaHack[36], Knee Osteoarthritis[37], MIAS[38], Mura[39], Pulmonary Chest MC[40], Pulmonary Chest Shenzhen[40], RUS CHN[41]), containing 5974 QA pairs of X-ray images.

Pathology validation datasets. Two pathology datasets were organized for evaluation: the test set of PathVQA[11] datset and the Omnimed_Pathology dataset. The test set of PathVQA dataset includes 6719 QA pairs related to pathology images. The Omnimed_Pathology dataset integrates 4 datasets (BreakHis[42], CRC100k[43], Malignant Lymph[44], NLM-Malaria Data[45]) and contains 2281 QA pairs.

Complex reasoning datasets. Four complex reasoning datasets were used for evaluation: MCQ_EN[18], MCQ_CN[18], MedMCQA_radiology[19] and MedMCQA_pathology[19] datasets. The MCQ_CN dataset contains 139 multi-choice questions designed for the Chinese ophthalmology senior professional title examination, and the MCQ_EN dataset is the translated English version of MCQ_CN dataset. The questions include six main topics: anterior segment diseases, external eye/orbital diseases, glaucoma, ocular trauma, refractive disorders/strabismus and retinal diseases. Each question contains 5 to 9 answer choices, with the number of correct answers ranging from 1 to 6. The MedMCQA_radiology and MedMCQA_pathology datasets are subsets of the MedMCQA[19] dataset, with 4395 and 14,884 real-world multiple-choice questions, respectively, designed for radiology and pathology medical entrance exams. The questions include both single-answer and multi-answer types.

Model design and training details

ConceptVLM employs pre-trained InternViT-300M[46] as the vision encoder. The InternViT-300M model consists of 0.3 billion parameters, 24 layers, a hidden size of 1024, and 16 attention heads. It employs standard LayerNorm without QK-Norm and is also pre-trained with the Next Token Prediction (NTP) loss. The language model used in ConceptVLM is the InternLM2-Chat-7B model[47], which has been trained on extensive multi-modal datasets such as ShareGPT4V[48], LLaVA-150K[49] and VQAv2[50]. InternLM2-Chat-7B is a 7-billion-parameter decoder-only Transformer, featuring Rotary Position Embeddings (RoPE) and Grouped Query Attention (GQA) to optimize memory efficiency while supporting 200K-token context windows. Each transformer layer features pre-layer normalization for stability and SwiGLU activation functions in feed-forward networks for enhanced nonlinear modeling. The model supports multi-image and video data alongside single-image and text-only data, enhancing its scalability for high-resolution processing.

Medical key concept dictionary. To accurately capture medical knowledge related to image-based questions while preventing the model from overemphasizing the form of responses, we have designed an algorithm for constructing a medical key concept dictionary[5,51,52]. For closed-end Q&A data, which typically require a specific word or phrase as an answer, such as identifying the modality of an image or categorizing a disease, we compile the closed-end key concept dictionary by directly tallying all unique answers across the dataset. This involves counting each distinct response and then removing duplicates, ensuring that the resulting dictionary encompasses all possible closed-form answers. Regarding open-end Q&A data, which can vary widely in format and content, we employ a length threshold criterion to distinguish between simple and complex responses. For answers containing fewer than K words, we construct an open-end key concept dictionary. This is motivated by the observation that very short open responses (e.g., "cystoid macular edema") are term-dense clinical phrases rather than free-form narratives, and thus benefit from focused supervision on high-value tokens. We set K as 5 based on parameter investigation experiments in Supplementary Table 4 on the PolyOphthalmology validation set. These shorter responses are likely to contain critical terms that can be effectively captured using our key concept-matching approach. Conversely, for responses exceeding the K-word threshold, indicating more complex or nuanced answers, we apply the traditional cross-entropy loss without additional processing. This allows the model to learn from diverse and detailed responses without imposing restrictive constraints.

Key concept-aware loss. Unlike conventional supervised fine-tuning which applies cross-entropy loss uniformly to all output tokens

regardless of semantic content, our method selectively supervises only those tokens that correspond to clinically salient concepts, effectively transforming the optimization objective from answer replication to diagnostic reasoning alignment. We first compute the overlap indices between the predicted tokens from the ConceptVLM, denoted as $T_{pred} \in R^{1\times n}$, and the dictionary tokens, denoted as $T_{dict} \in R^{1\times N}$. Here, T_{pred} represents the sequence of tokens generated by the ConceptVLM for a given input, with n being the total number of tokens in this sequence. Meanwhile, T_{dict} is the set of predefined medical key concepts in the dictionary, with N representing the total number of unique key concept tokens. Any token in T_{pred} that matches any token in T_{dict} is identified as a key concept that needs to be learned. Based on these matching indices, we construct a token mask $M \in R^{1\times n}$, where positions corresponding to non-matching tokens are set to 0, and positions corresponding to matching tokens are set to 1. The mask M serves to highlight the tokens that align with the medical key concepts, ensuring that only relevant tokens contribute to the loss computation. The loss calculation formula then incorporates this mask to selectively apply the cross-entropy loss only to the relevant tokens:

$$L_{kc} = -\sum_{i=1}^{n} M_i \cdot \sum_{c=1}^{C} y_{i,c} \log(y'_{i,c}) \tag{1}$$

In Eq. (1), L_{kc} denotes the key concept-aware loss. n is the length of the output token sequence. M_i is the mask value for the i-th token, which is 1 if the token matches a key concept and 0 otherwise. $y_{i,c}$ is the ground-truth indicator (1 if the true token at position i is vocabulary item c, 0 otherwise). C is the total number of unique tokens in the tokenizer. $y'_{i,c}$ is model's predicted probability for vocabulary item c at position i.

Dual loss strategy. Our training methodology introduces a distinctive approach to differentiate between closed-end and open-end question-and-answer (Q&A) data, alongside an innovative loss computation strategy. For closed-end Q&A data, where answers are generally short phrases or single words with a consistent format, we apply a key concept-aware loss function across all entries. On the other hand, open-end Q&A data is characterized by its varied formats and wide range of key concept categories, making it difficult to establish a comprehensive key concept dictionary for constraints. To tackle this issue, we introduce a length threshold criterion to decide between using traditional cross-entropy loss or the key concept-aware loss. If the length of the predicted answer is below the set threshold, indicating a concise response similar to those in closed-end questions, we use the key concept-aware loss. For longer and more complex answers typical of open-end questions, or when no predicted tokens match the medical key concept dictionary, we revert to conventional cross-entropy loss. The overall loss function is designed as follows:

$$L = \begin{cases} L_{kc} = -\sum_{i=1}^{n} M_i \cdot \sum_{c=1}^{C} y_{i,c} \log\left(y'_{i,c}\right), & \text{if closed or } \left(\text{open and } |y'| < K\right), \\ L_{ce} = -\sum_{i=1}^{n} \sum_{c=1}^{C} y_{i,c} \log\left(y'_{i,c}\right), & \text{otherwise.} \end{cases} \tag{2}$$

L_{kc} denotes the key concept-aware loss, and L_{ce} is the cross-entropy loss. n is the length of the output token sequence. C is the total number of unique tokens in the tokenizer. $|y'|$ represents the token length of the predicted answer. *closed* and *open* denote the type of current question.

Evaluation metrics and experimental settings

For the evaluation metrics on the SLAKE[10], PathVQA[11], and Poly-Ophthalmology datasets, we calculate accuracy for closed-ended questions and the ratio of ground-truth tokens appearing in generated sequences (Recall) for open-ended questions. The formulas are defined as follows:

$$Accuracy = \frac{number\ of\ correct\ predictions}{number\ of\ all\ samples} \times 100\% \tag{3}$$

$$Recall = \frac{\sum matched\ ground-truth\ tokens}{\sum total\ ground-truth\ tokens} \times 100\% \tag{4}$$

For the OmniMedVQA sub-datasets, we uniformly report the accuracy metric across all question types.

The model architecture is an extension of the Internvl-v2-8B foundational model, featuring a frozen backbone and LLM parameters, while keeping the MLP layers trainable. To ensure parameter-efficient tuning, a LoRA rank of 256 is applied to the LLM component. The length threshold K in medical key concept dictionary construction algorithm is set as 5. The training setup utilizes 2 GPUs, with each processing 4 samples per batch; gradients are accumulated over 8 steps to achieve an effective global batch size of 64. A dynamic resolution mechanism is employed to adaptively scale input images from a base size of 448 pixels, supporting up to 6 patch divisions and 50% downsampling, alongside thumbnail optimization to minimize computational overhead. Optimization is carried out using AdamW, with an initial learning rate of 1e-5, incorporating cosine decay scheduling, a 1% warmup phase, and a weight decay of 0.01. The training regimen covers 6 epochs, employing gradient checkpointing and limiting sequence lengths to 2048 tokens. The total training time was approximately 32 GPU-h, with a peak GPU memory consumption of approximately 69 GB.

We utilized Python version 3.9.20, alongside a suite of open-source libraries that enabled comprehensive experimentation and analysis. For the core of our training efforts, single NVIDIA A800 GPUs, each equipped with 80GB of VRAM, were employed. These GPUs were configured for multi-GPU training through DistributedDataParallel module (PyTorch version 2.2.1, CUDA 12.1), facilitating efficient parallel processing and significantly reducing training time. The underlying hardware also included an AMD EPYC 7713 64-Core Processor, providing robust computational support necessary for handling auxiliary tasks and data preprocessing. To manage large-scale model training more effectively, we incorporated DeepSpeed (version 0.13.5), which helped in optimizing memory usage and enabling the training of larger models than would otherwise be possible. Image input processing leveraged the Pillow library (version 11.0.0), ensuring high-quality image data handling. Additionally, TensorBoard (version 2.18.0) was used for visualization and monitoring of the training process, allowing us to track performance metrics and adjust parameters as needed. For specific functionalities within our pipeline, we employed Timm (version 0.9.12) for vision model architectures, torchvision (version 0.17.2) for additional image transformations and utilities, and OpenCV-Python (version 4.10.0.84) for advanced image processing tasks. This combination of hardware and software components created a powerful and flexible environment tailored to the demands of our ConceptVLM model training, enabling both efficiency and effectiveness in our research endeavors.

Statistical analysis

Two-sided t-tests were used to compare the accuracy and recall of ConceptVLM with those of other models. We used nonparametric bootstrapping to construct 95% confidence intervals (CI) calculated using 1.96 × standard error, sampling with the size of the test set. The P value was the proportion of differences in model performance greater than the observed difference in terms of absolute value. The null hypothesis was that there was no difference in model performance for the given test set and evaluation metric.

Ethics statement

This study was conducted in accordance with the Declaration of Helsinki and received approval from the Hong Kong Polytechnic University's institutional review board (HSEARS20240202004). The Institutional Review Board waived informed consent due to the retrospective analysis of anonymized ophthalmic images and public datasets.

Data availability

The datasets for constructing the PolyOphthalmology can be accessed via the links: ADAM (https://ieee-dataport.org/documents/adam-automatic-detection-challenge-age-related-macular-degeneration), APTOS-2019 (https://www.kaggle.com/competitions/aptos2019-blindness-detection/rules), BRSET (https://physionet.org/content/brazilian-ophthalmological/1.0.1/), DDR (https://github.com/nkicsl/DDR-dataset), DeepDRiD (https://github.com/deepdrdoc/DeepDRiD), drishtiGS (https://www.kaggle.com/datasets/lokeshsaipureddi/drishtigs-retina-dataset-for-onh-segmentation), EyePACS (https://www.kaggle.com/c/diabetic-retinopathy-detection), GAMMA (https://aistudio.baidu.com/competition/detail/119/0/introduction), JustRAIGS (https://justraigs.grand-challenge.org/dataset/), LAG (https://github.com/smilell/AG-CNN?tab=readme-ov-file), MESSIDOR2 (https://www.adcis.net/en/third-party/messidor2/), OCTDL (https://data.mendeley.com/datasets/sncdhf53xc/4), OCTID (https://www.openicpsr.org/openicpsr/project/108503/version/V1/view), ODIR (https://odir2019.grand-challenge.org/dataset/), ORIGA (https://service.tib.eu/ldmservice/dataset/origa-dataset), PALM (https://palm.grand-challenge.org/), PAPILA (https://figshare.com/articles/dataset/PAPILA/14798004/1), REFUGE (https://refuge.grand-challenge.org/), RFMID (https://ieee-dataport.org/open-access/retinal-fundus-multi-disease-image-dataset-rfmid), RIMONE (https://github.com/miag-ull/rim-one-dl), STARE (https://cecas.clemson.edu/~ahoover/stare/probing/index.html), Toxo (https://www.kaggle.com/datasets/nafin59/ocular-toxoplasmosis-fundus-images-dataset), Retina Image Bank (https://imagebank.asrs.org/). Downstream datasets can be accessed via the links: SLAKE (http://www.med-vqa.com/slake), PathVQA (https://github.com/UCSD-AI4H/PathVQA), ACRIMA (https://www.kaggle.com/datasets/toaharahmanratul/acrima-dataset), Adam Challenge (https://amd.grand-challenge.org/), Diabetic Retinopathy (https://tianchi.aliyun.com/dataset/93926), DRIMDB (https://www.kaggle.com/datasets/subhajournal/drimdb-diabetic-retinopathy-images-database), JSIEC (https://pmc.ncbi.nlm.nih.gov/articles/PMC8355164/#Sec16), OCT & X-Ray 2017 (https://data.mendeley.com/datasets/rscbjbr9sj/3), OLIVES (https://github.com/olivesgatech/OLIVES_Dataset), Yangxi (https://doi.org/10.5281/zenodo.3393265), Chest CT Scan (https://tianchi.aliyun.com/dataset/93929), Covid CT (https://tianchi.aliyun.com/dataset/106604), SARS-CoV-2 CT-scan (https://www.kaggle.com/datasets/plameneduardo/sarscov2-ctscan-dataset), Chest X-Ray PA (https://www.kaggle.com/datasets/prashant268/chest-xray-covid19-pneumonia), CoronaHack (https://github.com/ieee8023/covid-chestxray-dataset), Knee Osteoarthritis (https://www.kaggle.com/datasets/shashwatwork/knee-osteoarthritis-dataset-with-severity), MIAS (https://www.kaggle.com/datasets/kmader/mias-mammography), Mura (https://stanfordmlgroup.github.io/competitions/mura/), Pulmonary Chest MC (http://archive.nlm.nih.gov/repos/chestImages.php), Pulmonary Chest Shenzhen (http://archive.nlm.nih.gov/repos/chestImages.php), RUS CHN (https://aistudio.baidu.com/datasetdetail/69582/0), BreakHis (https://www.kaggle.com/datasets/ambarish/breakhis), CRC100k (https://paperswithcode.com/dataset/crc100k), Malignant Lymph (https://www.kaggle.com/datasets/andrewmvd/malignant-lymphoma-classification), NLM-Malaria Data (https://lhncbc.nlm.nih.gov/LHC-research/LHC-projects/image-processing/malaria-datasheet.html).

Code availability

Code available at https://github.com/Lewislou/conceptVLM/.

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Author contributions

D.S. and W.L. conceived the study. WL performed the experiments. M.H. provided data and computing facilities. Y.W. and P.X. manually labeled data. W.L. and Y.W. drafted the manuscript. W.Z., X.C., J.Y., M.H., and D.S. contributed to key data interpretation. All authors have critically revised the manuscript, read and approved the final version for submission.

Competing interests

The authors declare no competing interests.

Additional information

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